

Homework 3

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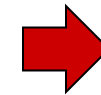
- Released in Blackboard
- Covers Chapters 8-10
- Due: 11:59pm, Apr. 30
- No late homework will be accepted!

▼ 算法设计与分析

主页

公告

Slides



Homework

Piazza

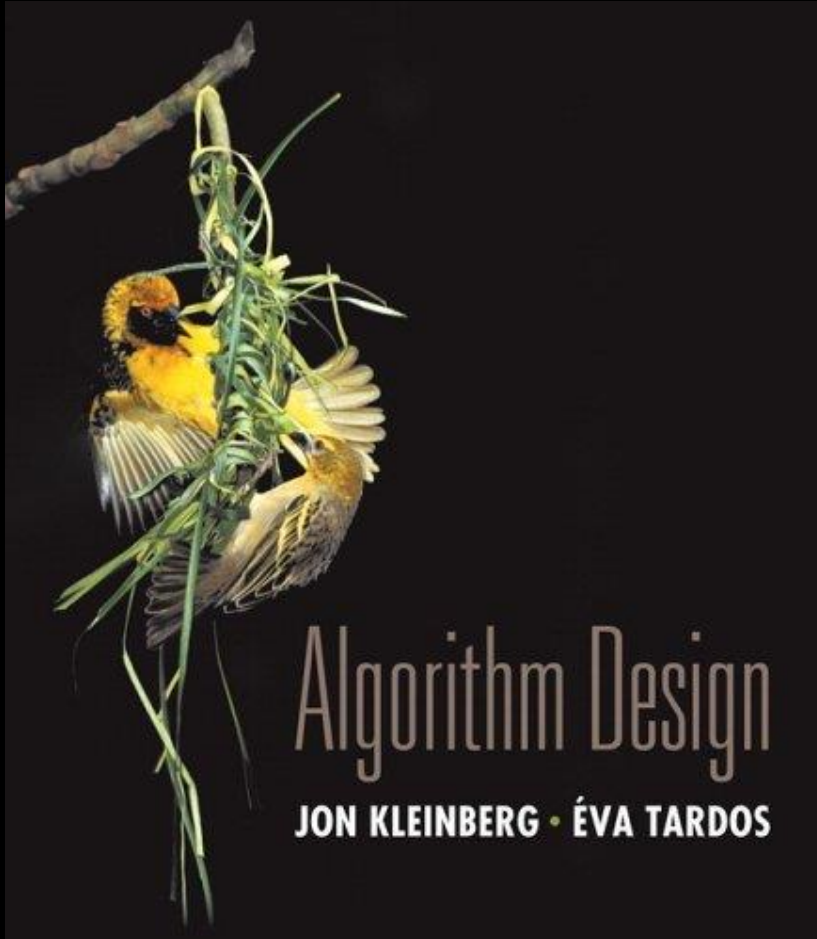
Course evaluation

<https://evaluation.shanghaitech.edu.cn/tqnmaep/login>



Chapter 9

PSPACE: A Class of Problems Beyond NP



Slides by Kevin Wayne.
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Geography Game

Geography. Amy names capital city c of country she is in. Bob names a capital city c' that starts with the letter on which c ends. Amy and Bob repeat this game until one player is unable to continue. Does Amy have a forced win?

Ex. Budapest $\rightarrow$ Tokyo $\rightarrow$ Ottawa $\rightarrow$ Ankara $\rightarrow$ Amsterdam $\rightarrow$ Moscow $\rightarrow$ Washington $\rightarrow$ Nairobi $\rightarrow$...

Geography on graphs. Given a directed graph $G = (V, E)$ and a start node s , two players alternate turns by following, if possible, an edge out of the current node to an **unvisited** node. Can first player guarantee to make the last legal move?

Remark. Some problems (especially involving 2-player games and AI) defy classification according to P, NP, and NP-complete.

9.1 PSPACE

PSPACE

P. Decision problems solvable in polynomial **time**.

PSPACE. Decision problems solvable in polynomial **space**.

Observation. $P \subseteq PSPACE$.

↑
poly-time algorithm can consume only polynomial space

PSPACE

Claim. 3-SAT is in PSPACE.

Pf.

- Enumerate all 2^n possible truth assignments.
- Check each assignment to see if it satisfies all clauses. ▀

Theorem. $NP \subseteq PSPACE$.

Pf. Consider arbitrary problem Y in NP.

- Since $Y \leq_p 3\text{-SAT}$, there exists algorithm that solves Y in poly-time plus polynomial number of calls to 3-SAT black box.
- Can implement black box in poly-space. ▀

9.3 Quantified Satisfiability

Quantified Satisfiability

QSAT (Quantified 3-SAT). Let $\Phi(x_1, \dots, x_n)$ be a Boolean CNF formula. Is the following propositional formula true?

$$\exists x_1 \forall x_2 \exists x_3 \forall x_4 \dots \forall x_{n-1} \exists x_n \Phi(x_1, \dots, x_n)$$

↑
assume n is odd

Intuition. Amy picks truth value for x_1 , then Bob for x_2 , then Amy for x_3 , and so on. Can Amy satisfy Φ no matter what Bob does?

Ex. $(x_1 \vee x_2) \wedge (x_2 \vee \overline{x_3}) \wedge (\overline{x_1} \vee \overline{x_2} \vee x_3)$

Yes. Amy sets x_1 true; Bob sets x_2 ; Amy sets x_3 to be same as x_2 .

Ex. $(x_1 \vee x_2) \wedge (\overline{x_2} \vee \overline{x_3}) \wedge (\overline{x_1} \vee \overline{x_2} \vee x_3)$

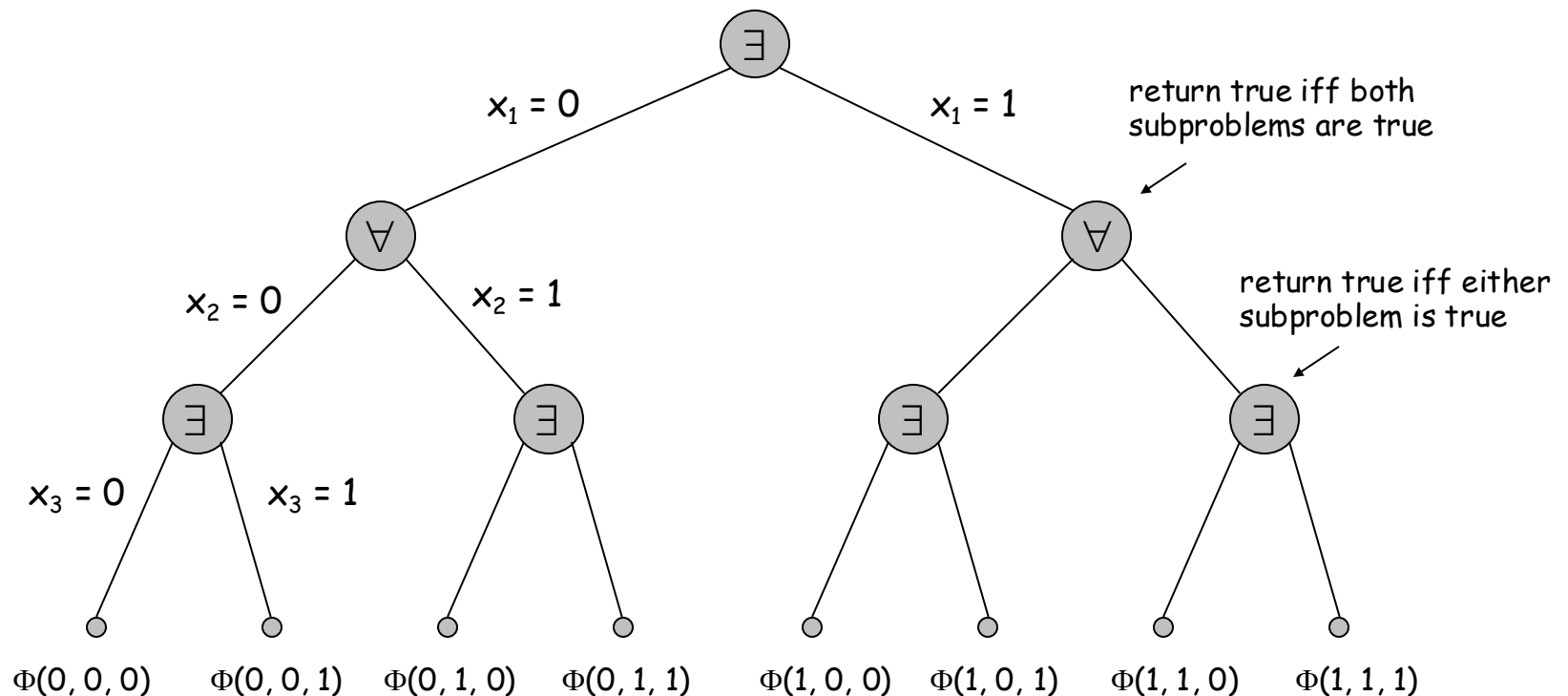
No. If Amy sets x_1 false; Bob sets x_2 false; Amy loses;
if Amy sets x_1 true; Bob sets x_2 true; Amy loses.

QSAT is in PSPACE

Theorem. $QSAT \in PSPACE$.

Pf. Recursively try all possibilities.

- Only need one **bit** of information from each subproblem.
- Amount of space is proportional to **depth** of function call stack.

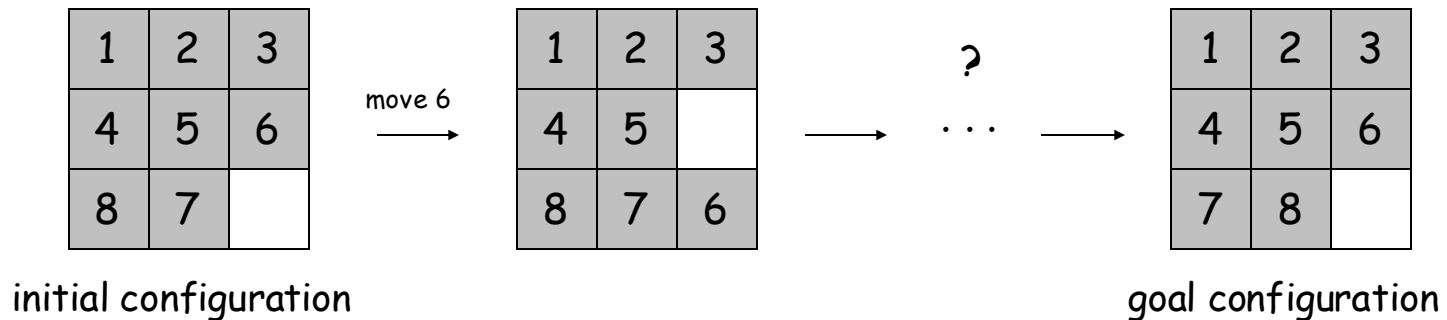


9.4 Planning Problem

15-Puzzle

8-puzzle, 15-puzzle. [Sam Loyd 1870s]

- Board: 3-by-3 grid of tiles labeled 1-8.
- Legal move: slide neighboring tile into blank (white) square.
- Find sequence of legal moves to transform initial configuration into goal configuration.



Planning Problem

Conditions. Set $C = \{ C_1, \dots, C_n \}$.

Initial configuration. Subset $c_0 \subseteq C$ of conditions initially satisfied.

Goal configuration. Subset $c^* \subseteq C$ of conditions we seek to satisfy.

Operators. Set $O = \{ O_1, \dots, O_k \}$.

- To invoke operator O_i , must satisfy certain prereq conditions.
- After invoking O_i certain conditions become true, and certain conditions become false.

PLANNING. Is it possible to apply sequence of operators to get from initial configuration to goal configuration?

Examples.

- 15-puzzle.
- Rubik's cube.
- Logistical operations to move people, equipment, and materials.

Planning Problem: 8-Puzzle

Planning example. Can we solve the 8-puzzle?

Conditions. $C_{ij}, 1 \leq i, j \leq 9$. $\leftarrow C_{ij}$ means tile i is in square j

Initial state. $c_0 = \{C_{11}, C_{22}, \dots, C_{66}, C_{78}, C_{87}, C_{99}\}$.

Goal state. $c^* = \{C_{11}, C_{22}, \dots, C_{66}, C_{77}, C_{88}, C_{99}\}$.

Operators.

- Precondition to apply $O_i = \{C_{78}, C_{99}\}$.
- After invoking O_i , conditions C_{79} and C_{98} become true.
- After invoking O_i , conditions C_{78} and C_{99} become false.

1	2	3
4	5	6
8	7	9

↓ O_i

1	2	3
4	5	6
8	9	7

(This is the most straightforward cast of 8-puzzle to planning, though better cast exists.)

Diversion: Why is 8-Puzzle Unsolvable?

8-puzzle invariant. Any legal move preserves the **parity** of the number of pairs of pieces in reverse order (inversions).

3	1	2
4	5	6
8	7	

3 inversions
1-3, 2-3, 7-8



3	1	2
4	5	6
8		7

3 inversions
1-3, 2-3, 7-8



3	1	2
4		6
8	5	7

5 inversions
1-3, 2-3, 7-8, 5-8, 5-6

1	2	3
4	5	6
7	8	

0 inversions



1	2	3
4	5	6
8	7	

1 inversion: 7-8

Planning Problem: Binary Counter

Planning example. Can we increment an n -bit counter from the all-zeroes state to the all-ones state?

Conditions. $C_1, \dots, C_n$. $\leftarrow C_i$ corresponds to bit $i = 1$

Initial state. $c_0 = \phi$. $\leftarrow$ all 0s

Goal state. $c^* = \{C_1, \dots, C_n\}$. $\leftarrow$ all 1s

Operators. $O_1, \dots, O_n$.

- To invoke operator O_i , must satisfy $C_1, \dots, C_{i-1}$. $\leftarrow$ $i-1$ least significant bits are 1
- After invoking O_i , condition C_i becomes true. $\leftarrow$ set bit i to 1
- After invoking O_i , conditions $C_1, \dots, C_{i-1}$ become false. $\leftarrow$ set $i-1$ least significant bits to 0

Solution. $\{\} \Rightarrow \{C_1\} \Rightarrow \{C_2\} \Rightarrow \{C_1, C_2\} \Rightarrow \{C_3\} \Rightarrow \{C_3, C_1\} \Rightarrow \dots$

Observation. Any solution requires at least $2^n - 1$ steps.

Planning Problem: In Exponential Time

Configuration graph G .

- Include node for each of 2^n possible configurations.
- Include an edge from configuration c' to configuration c'' if one of the operators can convert from c' to c'' .

PLANNING. Is there a path from c_0 to c^* in configuration graph?

Claim. PLANNING is in EXPTIME.

Pf. Run BFS to find path from c_0 to c^* in configuration graph. ▀

Note. Configuration graph can have 2^n nodes, and shortest path can be of length $= 2^n - 1$.

Planning Problem: In Polynomial Space

Theorem. PLANNING is in PSPACE.

Pf.

- Suppose there is a path from c_1 to c_2 of length $\leq L$.
- Path from c_1 to midpoint and from midpoint to c_2 are each $\leq L/2$.
- Enumerate all possible midpoints.
- Apply recursively. Depth of recursion = $\log_2 L$. ▫

```
boolean hasPath( $c_1$ ,  $c_2$ , L) {  
    if ( $L = 1$ ) return correct answer  
  
    foreach configuration  $c'$  {  
        boolean  $x = \text{hasPath}(c_1, c', \lceil L/2 \rceil)$   
        boolean  $y = \text{hasPath}(c', c_2, \lfloor L/2 \rfloor)$   
        if ( $x$  and  $y$ ) return true  
    }  
    return false  
}
```

9.5 PSPACE-Complete

PSPACE-Complete

PSPACE. Decision problems solvable in polynomial space.

PSPACE-Complete. Problem Y is PSPACE-complete if (i) Y is in PSPACE and (ii) for every problem X in PSPACE, $X \leq_p Y$.

Theorem. [Stockmeyer-Meyer 1973] QSAT is PSPACE-complete.

Theorem. $\text{PSPACE} \subseteq \text{EXPTIME}$.

Pf. Previous algorithm solves QSAT in exponential time, and QSAT is PSPACE-complete. ▀

Summary. $P \subseteq NP \subseteq \text{PSPACE} \subseteq \text{EXPTIME}$.

↑ ↑ ↑
it is known that $P \neq \text{EXPTIME}$, but unknown which inclusion is strict;
conjectured that all are

PSPACE-Complete Problems

More PSPACE-complete problems.

- ▣ **Competitive facility location.**
- ▣ Natural generalizations of games.
 - Othello, Hex, Geography, Rush-Hour, Instant Insanity
 - Shanghai, go-moku, Sokoban
- ▣ Given a memory restricted Turing machine, does it terminate in at most k steps?
- ▣ Do two regular expressions describe different languages?
- ▣ Is it possible to move and rotate complicated object with attachments through an irregularly shaped corridor?
- ▣ Is a deadlock state possible within a system of communicating processors?

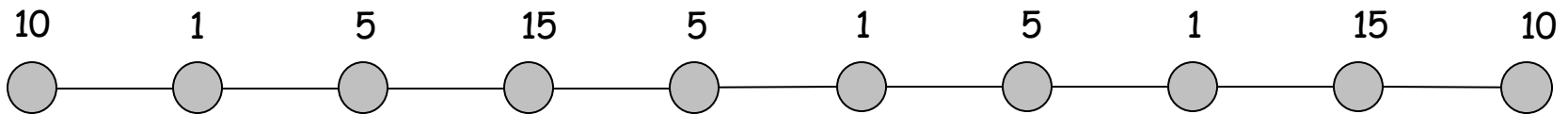
Competitive Facility Location

Input. Graph with positive node weights, and target number B .

Game. Two competing players alternate in selecting nodes. Not allowed to select a node if any of its neighbors has been selected.

Competitive facility location. Can **second** player guarantee at least B units of profit?

(In general, the graph may not be a chain.)



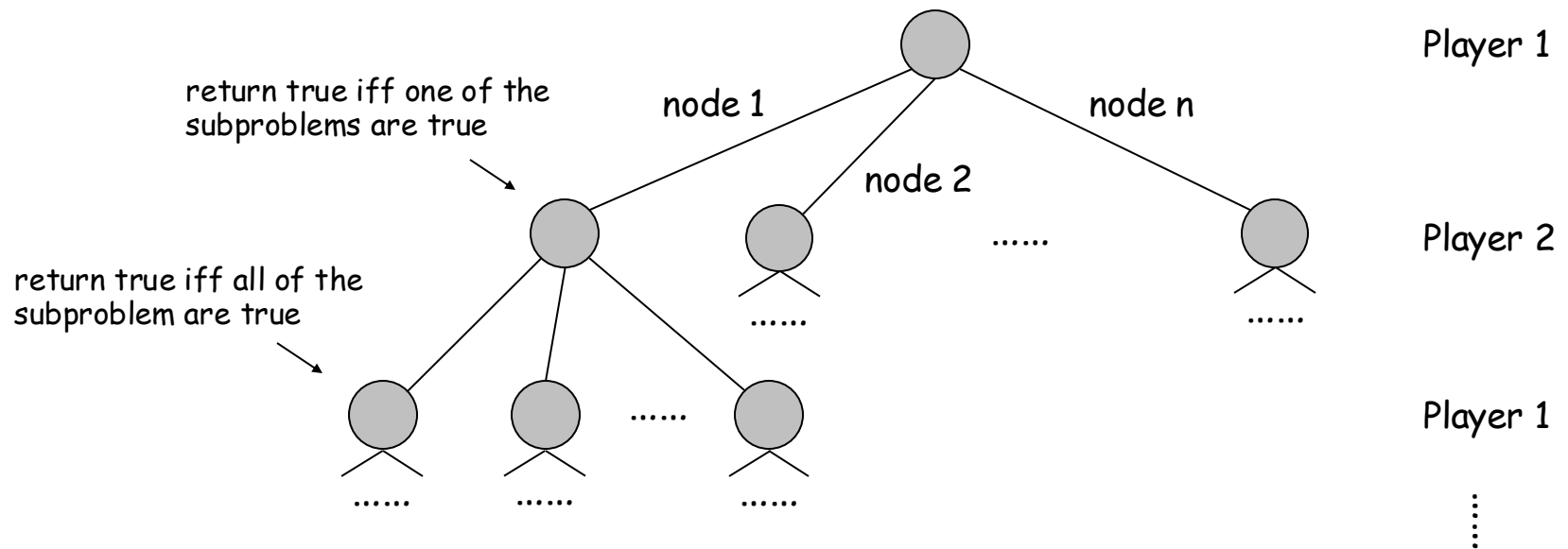
Yes if $B = 20$; no if $B = 25$.

Competitive Facility Location

Claim. COMPETITIVE-FACILITY is PSPACE-complete.

Pf.

- To solve it in poly-space, use recursion like QSAT, but at each step there are up to n choices instead of 2.



Competitive Facility Location

Claim. COMPETITIVE-FACILITY is PSPACE-complete.

Pf.

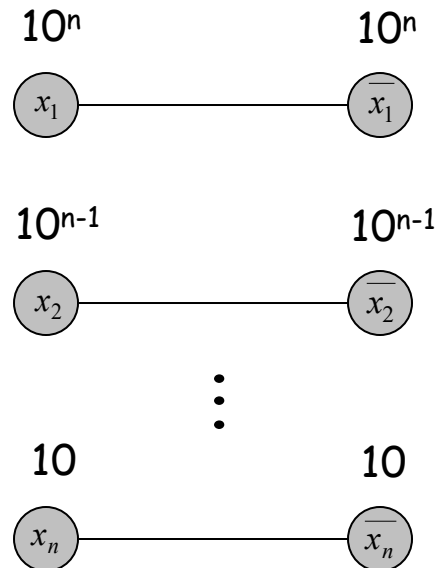
- To show that it's complete, we show that QSAT polynomial reduces to it. Given an instance of QSAT, we construct an instance of COMPETITIVE-FACILITY such that player 2 can force a win iff QSAT formula is false.

Competitive Facility Location

- Construction.** Given instance $\Phi(x_1, \dots, x_n) = C_1 \wedge C_2 \wedge \dots \wedge C_k$ of QSAT. ← assume n is odd
- Include a node for each literal and its negation and connect them.
 - at most one of x_i and its negation can be chosen
 - Choose a large constant c (e.g., $c \geq k+2$), and put weight c^{n-i+1} on literal x_i and its negation;

set $B = c^{n-1} + c^{n-3} + \dots + c^4 + c^2 + 1$.

 - This ensures variables are selected in order $x_1, x_2, \dots, x_n$.
 - As is, player 2 will lose by 1 unit: $c^{n-1} + c^{n-3} + \dots + c^4 + c^2$.



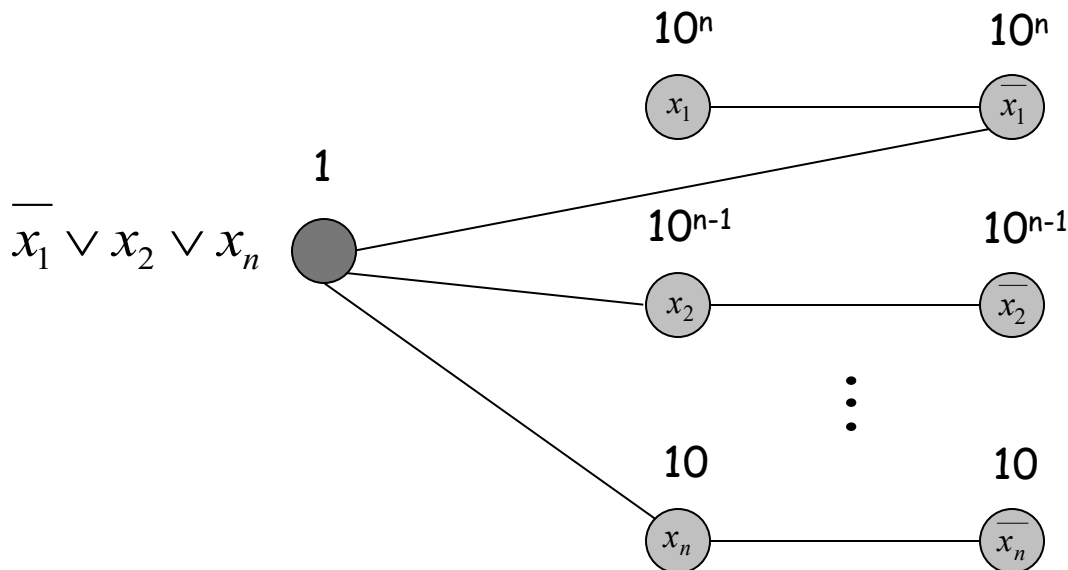
Competitive Facility Location

Construction. Given instance $\Phi(x_1, \dots, x_n) = C_1 \wedge C_2 \wedge \dots \wedge C_k$ of QSAT.

- Give player 2 **one last move** on which she can try to win.
- For each clause C_j , add node with value 1 and an edge to each of its literals.
- Player 2 can make last move iff truth assignment defined alternately by the players failed to satisfy some clause, i.e.,

$$\forall x_1 \exists x_2 \forall x_3 \exists x_4 \dots \exists x_{n-1} \forall x_n \neg \Phi(x_1, \dots, x_n)$$

$$\Leftrightarrow \neg \exists x_1 \forall x_2 \exists x_3 \forall x_4 \dots \forall x_{n-1} \exists x_n \Phi(x_1, \dots, x_n)$$



Chapter Summary

PSPACE

PSPACE. Decision problems solvable in polynomial **space**.

PSPACE problems

- QSAT
- Planning

Theorem. $NP \subseteq PSPACE \subseteq EXPTIME$

PSPACE-Complete. Problem Y is PSPACE-complete if (i) Y is in PSPACE and (ii) for every problem X in PSPACE, $X \leq_p Y$.

PSPACE-Complete problems

- QSAT
- Competitive Facility Location

Complexity Classes

